Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/18/2021
PI Name:	Delp, Michael Ph.D.		
Project Title:	Effects of Microgravity on Cerebral Arterial, Venous, and Lymphatic Function: Implications for Elevated Intracranial Pressure		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
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No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS Rodent Research-9 NOTE: Extended to 11/30/2019 per NSSC information (Ed., 12/11/18)		
Key Personnel Changes/Previous PI:	March 2017: No changes.		
COI Name (Institution):	Zawieja, David Ph.D., M.D. (Texas A&M University)		
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Task Description:	NOTE: Continuation of "Effects of Microgravity on Cerebral Arterial, Venous, and Lymphatic Function: Implications for Elevated Intracranial Pressure," grant #NNX13AN33G, due to PI move to Florida State University. Approximately 29% of astronauts on short-term (~2 wk) space shuttle flights and 60% on long-duration (~6 mo) missions to the International Space Station (ISS) are reported to have experienced some impairment in distant or near visual acuity. These visual disturbances have been hypothesized to be related to increases in intracranial pressure. Increases in intracranial pressure associated with spaceflight are thought to occur through the cephalad shift of intravascular and cerebrospinal fluids due to the loss of the head-to-foot gravity vector present on Earth. In regard to the vascular component, increases in cerebral blood flow and capillary fluid filtration could elevate fluid movement into the cranium, while impaired function of cerebral veins and lymphatics could impede intracranial fluid drainage out of the cranium. Therefore, the purpose of this proposal is to examine the effects of long-term spaceflight on the vasoconstrictor, mechanical, and structural properties of cerebral arteries, veins, and lymphatic vessels. Flight: The aim of this project is to investigate whether spaceflight on the ISS alters the blood-brain barrier in rodents, as indicated by ultrastructural examination of the cerebral capillary endothelium. To assess the changes that occur in the structure of the blood-brain barrier, 10-15 female C57BL/6 mice, 10-11 weeks old will be loaded into a transporter (Animal Enclosure Module AEM-T; 10 mice/side), and late-loaded into a SpaceX Dragon rocket. After ascent and docking to the ISS, the mice will be transferred via an Animal Access Unit (AAU) and Mouse Transfer Box (MTB) to 2 Habitats (AEM-X). The mice will be grouped into four experimental groups, each housed in one side of an AEM-X. The groups, as determined by another investigation, will consist of mice receiving intraperitoneal (i.p.) injection of saline vehicle (n=5), Tetanus Toxoid (TT) vaccine (n=5), CpG ODN adjuvant (n=5), or TT vaccine + CpG ODN adjuvant (n=5). After mice have been exposed to microgravity for a minimum of 2 weeks, animals for this study will be inoculated with an injection of phosphate buffered saline (n=5) or CpG ODN adjuvant (n=5); a third group of animals injected with the TT vaccine (n=5) may be utilized. Two weeks later (14 +/- 2 days), the mice will be sacrificed in the Materials Science Glovebox (MSG). Sacrificing will be achieved via i.p. injection of 100/10 mg/kg ketamine/xylazine anesthesia followed by exsanguination by cardiac puncture. Following blood collection and harvest of spleen for another research investigation, the animals will have heads removed. The top of the skull will be cut along the sagittal crest and suture and retracted back to expose the brain. The brains will be will be dissected from the mice and put in a solution of 10% buffered formalin at ambient temperature. 24-48 hours later, the 10% buffered formalin will be exchanged with the Brain Fixation Kit (BFK) to 70% Ethanol and stored at ambient temperature until they can be returned to Earth and analyzed. Ground: 20-25 (saline vehicle (n=10), Tetanus Toxoid (TT) vaccine (n=5), CpG ODN adjuvant (n=10)) ground control mice will be put in AEMs inside Orbiter Environmental Simulators (OES) on a 2-8 day delay, depending on the when telemetry data from the SpaceX Dragon and ISS environments are available. The differences in vascular structure between ground controls and flight animals with a P-value < 0.05 will be considered statistically significant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The lymphatic system is one of the least understood parts of the circulatory system in the body. This project will be the first to examine the effects of microgravity on lymphatic function. On Earth, understanding gained from these studies will help us better understand the adaptability of the lymphatic circulation and how it might respond to chronic bedrest and physical deconditioning. These studies may also provide insight into intracranial hypertension that occurs on Earth.
	[Ed. note: Compiled from final report submitted October 2020 to NASA Ames Research Center] Experimental Approach: Male mice were studied as part of the Rodent Research 9 (RR-9) mission, which was launched on SpaceX-12 in August, 2017. The animals were flown to the International Space Station (ISS) where they remained for approximately 33 days before being returned to Earth. The vascular physiology experiments with the Flight animals and a group of Vivarium Control mice were conducted at Loma Linda University in Loma Linda, California, in October, 2017. Habitat Ground Control (HC) and Vivarium Control (VC) studies were conducted at Kennedy Space Center (KSC) in Cape Canaveral, Florida in July, 2018. Results: Aim 1. To determine whether spaceflight on the ISS alters rodent basilar artery spontaneous tone, myogenic and KCl-evoked vasoconstriction, mechanical stiffness and gross structure. Histological measurements of basilar artery gross vascular structure (e.g., medial wall thickness and intraluminal diameter) did not show significant differences between RR-9 flight mice (FLT) and control mice. However, basilar arteries from the FLT mice had lower smooth muscle cell (SM22a) and endothelial cell (CD31) expression relative to that in controls, indicating the proportional composition of the basilar artery was altered by spaceflight. These changes in the proportional composition of the basilar artery did not alter the passive mechanical behavior or stiffness of the artery. The changes in the structural composition of the basilar artery did, however, appear to have functional consequences to both vasoconstrictor and vasodilator responsiveness. Similar to the results previously found in mouse basilar arteries flown on the Space Shuttle STS-135 and BionM1 missions, we found that basilar arteries in the RR-9 flight mice showed diminished vasoconstrictor responses to KCl, a non-receptor agonist, and U46619, a thromboxane A2 receptor agonist, relative to that in the Habitat Control mice (Hab) and Vivarium Control mice (FViv+KSCViv). This reduction in both non-receptor and receptor-mediated vasoconstriction is associated with the diminished expression of smooth muscle cells in the basilar artery. Vasodilation mediated through basilar artery endothelial cells via acetylcholine was diminished in the Spaceflight (SF) mice relative to that in the Habitat Control mice (HC) and Vivarium Control mice (VC). These results are in agreement with results obtained with mice flown on the Bion-M1 mission, where a single maximal dose of acetylcholine showed lower endotheliummediated vasodilation in SF mice relative to that in HC and VC mice. Smooth muscle, endothelium-independent vasodilation via the nitric oxide donor diethylamine NONOate diethylammonium (DEA/NO) was also lower in basilar arteries from the SF mice compared to control animals. These data indicate that both endothelium-dependent and -independent vasodilation of cerebral arteries are compromised by spaceflight. To examine possible mechanisms for the impaired vasodilator responses, we used immunofluorescence to examine the

Task Progress:	expression of basilar artery: 1) endothelial nitric oxide synthase (eNOS) expression, the enzyme responsible for generating the vasodilator molecule nitric oxide, 2) superoxide dismutase (SOD) expression, an anti-oxidant enzyme, and 3) prooxidant enzyme expression of NADPH oxidase 2 (NOX2) and xanthine oxidase (XO). The results indicated that eNOS expression in the basilar artery did not differ between FLT and control mice. However, expression of the anti-oxidant enzyme SOD was lower in the FLT mice and expression of the pro-oxidant enzyme XO was higher; NOX2 expression was not different between groups. These data indicate that spaceflight elevates oxidative stress in cerebral arteries through reductions in anti-oxidant protection and increases in production of reactive oxygen species. Aim 2. To estimate whether spaceflight on the ISS alters the blood-brain barrier (BBB) in rodents, as indicated by ultrastructural examination of the junctional complex of the cerebral capillary endothelium. Two astrocyte biomarkers were used to examine the integrity and function of the BBB and microvessel damage: glial fibrillary acidic protein (GFAP) and aquaporin-4 (AQP4). GFAP is present in the soma and astrocytic process and serves as a target protein for immunohistochemical detection of astrocytes. AQP4 is a water channel protein concentrated at the luminal surfaces of astrocyte end-feet, which outline the vascular bed to which they attach. Immunohistochemical analysis of the mouse brain revealed a significant increase in expression of AQP4 in the Flight mice (FLT) compared to that in Ground Control mice (GC). Tissue expression of ZO-1 and PECAM-1 in the brain were evaluated. There was a significant decrease in the expression of the BBB-related tight junction protein ZO-1 and an increase in the expression of PECAM-1. Like those of AQP4, these results also indicate a disturbance in the integrity of the BBB. Using immunohistochemical methods to determine the number of TUNEL-positive cells in the brain, quantitative assessment of brain sections demonstrated that spaceflight caused a 2.2-fold increase in apoptotic cells in the hippocampus of FLT mice compared to that in GC groups. Collectively, these results demonstrate spaceflight-induced apoptosis in the brain (TUNELpositive cells) and evoked changes in the expression of BBB-related proteins, AQP-4, ZO-1 and PECAM-1, indicate a disruption in the integrity and compromised function of the BBB. Aim 3. To determine whether spaceflight on the ISS alters rodent sagittal sinus vein (inside cranium) and jugular vein (outside cranium) spontaneous tone, myogenic and NE-evoked constriction, distension and gross structure. Using in vitro methods, the maximal passive diameter (4 cmH2O) of the jugular vein was determined in RR-9 Flight, Habitat Control, and Vivarium control mice. The maximal diameter was smaller in Flight mice relative to that in the control groups. Additionally, jugular wall thickness was thinner in FLT mice relative to that in Control animals. The combination of both a decrease in maximal diameter and a thinning of the medial smooth muscle layer in the jugular vein of FLT mice resulted in a reduction in the intima-media cross-sectional area. Using immunofluorescent histology to determine the smooth muscle cell (SM22a) and endothelial cell (CD31) composition of the jugular vein, the results indicate a non-significant decrease in SM22a expression and a significant decrease in CD31 expression in jugular veins from FLT mice relative to that in Controls, indicating the composition of the jugular was altered by spaceflight. Collectively, such changes in the gross structural characteristics and composition of the jugular vein could have functional consequences on vein function and ultimately blood flow through the vessel. In vitro studies were conducted to determine the effects of spaceflight on the functional vasoconstrictor properties of the jugular vein. The data suggest that spaceflight enhances the venoconstrictor response of jugular veins to norepinephrine as well as myogenic venoconstrictor tone across a range of intravascular pressures. Such adaptations could serve to promote venous congestion in the head and contribute to possible elevations in intracranial pressure. To determine whether spaceflight-induced reductions in endothelial cell release of nitric oxide or alterations in reactive oxygen species might be associated with the increase in venoconstrictor tone of the jugular veins, we used immunofluorescence to examine the expression of eNOS, SOD, NOX2, and XO in the jugular vein. The results indicated a reduction of eNOS expression in the jugular veins of FLT mice, but no difference in the expression of the anti-oxidant enzyme SOD or the expression of the pro-oxidant enzymes NOX2 and XO. These results suggest the enhanced venoconstrictor tone of the jugular veins from FLT mice could be in part the result of fewer endothelial cells lining the jugular vein and diminished basal release of NO through the eNOS signaling pathway. Finally, we did not find any differences in the maximal diameter or markers of smooth muscle cell (SM22a) and endothelial cell (CD31) composition of the sagittal sinus vein between FLT and Control mice. These results indicate that venous vessel adaptations to spaceflight may be different within and outside the cranium. Aim 4. To determine whether spaceflight on the ISS alters the ability of the cervical lymphatics modulate lymph flow, and thus, regulate cerebral fluid homeostasis. In vitro studies of isolated cervical pre-nodal lymphatics were conducted to determine whether spaceflight alters the functional properties of lymphatic vessels of the head and neck. The data indicate that spaceflight increases the passive diameter of cervical lymphatics by 10-15%, increases vessel tone by approximately 50%, elevates the pumping frequency by approximately 75%, and overall results in an increase in pump function by approximately 20-25%. Mechanistically, the overall elevation in pump function may be the result of alterations in the flow/shear-dependent responses of cervical lymphatic vessels, which can serve to enhance pump frequency, amplitude and flow in lymphatic vessels. In addition to cervical lymphatic vessels, we examined the gross structural properties of meningeal lymphatics, which unlike the cervical lymphatics of the neck, reside within the cranium. We found both an increase in wall thickness and maximal diameter of meningeal lymphatic vessels. These data suggest that both pressure and flow within the intracranial lymphatics are elevated with spaceflight.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Mao XW, Nishiyama NC, Byrum SD, Stanbouly S, Jones T, Holley J, Sridharan V, Boerma M, Tackett AJ, Willey JS, Pecaut MJ, Delp MD. "Spaceflight induces oxidative damage to blood-brain barrier integrity in a mouse model." FASEB J. 2020 Nov;34(11):15516-30. Epub 2020 Sep 26. https://doi.org/10.1096/fj.202001754R ; PMID: 32981077; PMCID: PMC8191453 , Nov-2020

Articles in Peer-reviewed Journals	Willey JS, Britten RA, Blaber E, Tahimic CGT, Chancellor J, Mortreux M, Sanford LD, Kubik AJ, Delp MD, Mao XW. "The individual and combined effects of spaceflight radiation and microgravity on biologic systems and functional outcomes." J Environ Sci Health C Toxicol Carcinog. 2021 Apr 27;39(2):129-79. https://doi.org/10.1080/26896583.2021.1885283 ; PMID: 33902391 , Apr-2021